

H2 Economics Case Study Practice 3

Standard of Living — Short-Answer Questions (4–6 marks)

H2 Economics 9570 | Compiled by Jeremy Lim

Instructions to students. This booklet contains three case studies on standard of living. Case Study 1 compares living standards across two large economies, the United States and China, and asks whether GDP per capita is a meaningful basis for cross-country comparison. Case Study 2 examines living standards and income inequality in Singapore. Case Study 3 considers wider measures of living standards beyond GDP, drawing on the Human Development Index and Bhutan’s Gross National Happiness framework. Each case study contains an extract and a data table, followed by six short-answer questions ranging from 4 to 6 marks. Write your answers in the lined spaces provided. Use the mark allocation as a guide to depth: 4-mark questions typically require a two-step explanation; 6-mark questions require a developed analytical response with reference to the data or relevant economic theory.

CASE STUDY 1. Comparing Living Standards: The United States and China

Table 1.1: Selected Economic and Social Indicators, United States and China, 2022

Indicator	United States	China
GDP per capita, current US\$	76,330	12,720
GDP per capita, PPP-adjusted US\$	76,330	21,480
Gini Coefficient (0–100, higher = more unequal)	39.7	38.2
Life Expectancy at Birth (years)	77.4	78.2
Adult Literacy Rate (%)	99.0	96.8
Mean Years of Schooling	13.7	7.6
PM2.5 Air Pollution ($\mu\text{g}/\text{m}^3$, annual mean)	7.4	38.0
Working Hours per Week (manufacturing)	41.5	49.0
Human Development Index (rank/191)	0.927 (21st)	0.768 (79th)

Source: Adapted from World Bank, OECD, UNDP Human Development Report 2023, and IMF World Economic Outlook.

Extract 1.1: Headline numbers versus lived reality

In 2022, the United States produced an estimated 76,330 US dollars of output per person, almost exactly six times the figure for China at 12,720 US dollars. On paper, this suggests that the average American enjoys a material standard of living six times higher than the average Chinese citizen. Yet most economists are reluctant to draw such a strong conclusion from a single number.

The first complication is exchange rates. When GDP per capita is adjusted for purchasing power parity (PPP), which corrects for the fact that non-traded goods such as rent, haircuts, and street food are typically cheaper in lower-income economies, China’s figure rises to 21,480 US dollars while the United States figure is unchanged. The PPP-adjusted ratio narrows to roughly 3.6 times, not 6.0 times.

A second complication is distribution. Both countries have similar Gini coefficients of around 38–40, but the composition of inequality differs. In the United States, top-income households capture a disproportionate share of national income. In China, the urban-rural gap remains large, with average disposable income in Beijing and Shanghai roughly three times that in interior rural provinces.

A third complication is non-monetary quality of life. The data show that China’s life expectancy now slightly exceeds that of the United States, while air quality is markedly worse. Working hours are longer in Chinese manufacturing, leaving less leisure time. Mean years of schooling are significantly higher in the United States, suggesting deeper human capital. The Human Development Index, which combines income, health, and education into a single composite, ranks the United States 21st and China 79th globally, a narrower gap than nominal GDP per capita would suggest.

Source: Adapted from Financial Times, “How not to compare living standards across borders”, 18 March 2024.

Question 1

[4 marks]

Define “standard of living”. Explain why GDP per capita is commonly used as a proxy for the material standard of living.

Question 2

[4 marks]

Using Table 1.1 and Extract 1.1, explain why the use of purchasing power parity (PPP) adjustment narrows the apparent income gap between the United States and China.

Question 3

[4 marks]

Explain how income inequality, as measured by the Gini coefficient, complicates the use of GDP per capita as an indicator of the average citizen's standard of living.

Question 4

[6 marks]

With reference to Table 1.1, explain two non-monetary factors that limit the conclusion that the standard of living in the United States is six times higher than that in China.

Question 5

[6 marks]

Explain how the Human Development Index addresses some, but not all, of the limitations of using GDP per capita as a measure of the standard of living.

Question 6

[6 marks]

Explain two reasons why, despite its limitations, GDP per capita remains the most widely used indicator of the standard of living in cross-country comparisons.

CASE STUDY 2. Growth, Inequality, and Living Standards in Singapore

Table 2.1: Selected Singapore Living Standards Indicators, 2010–2023

Indicator	2010	2015	2019	2022	2023
GDP per capita, current US\$	47,237	55,646	65,833	82,808	84,734
Real Median Household Income, growth (% p.a.)	1.9	4.4	1.0	-0.2	2.8
Gini Coefficient, after transfers (0–1)	0.452	0.409	0.398	0.378	0.371
Resident Unemployment Rate (%)	2.2	2.8	2.9	2.6	2.8
Home Ownership Rate (%)	87.2	90.8	90.4	89.3	89.7
Life Expectancy at Birth (years)	81.7	82.7	83.6	83.0	83.5
Hours Worked per Week (resident, median)	46.0	45.3	44.7	42.6	42.4
Doctors per 1,000 population	1.7	2.2	2.5	2.7	2.8

Source: Adapted from Department of Statistics Singapore, Ministry of Manpower, and Ministry of Health.

Extract 2.1: Has growth translated into well-being?

Over the past decade, Singapore's GDP per capita has risen by nearly 80 per cent in nominal US dollar terms, propelling the country into the top tier of high-income economies. The headline figure, however, conceals a more nuanced picture of well-being. Real median household income from work has grown more modestly, and in 2022 contracted slightly due to high inflation eroding nominal wage gains.

The Gini coefficient after government transfers and taxes has declined steadily, falling from 0.452 in 2010 to 0.371 in 2023. This reflects a series of redistributive policies including the Workfare Income Supplement, the Progressive Wage Model in low-wage sectors, GST vouchers, and CDC vouchers. Government transfers and subsidies now reduce the pre-tax Gini coefficient by around eight to nine points, a larger reduction than in most ASEAN economies.

Beyond income, several non-monetary indicators have improved. Life expectancy has risen by nearly two years over the decade. Median weekly working hours have fallen, indicating greater leisure. The doctor-to-population ratio has increased, suggesting better access to healthcare. Home ownership remains near 90 per cent, one of the highest rates in the world.

Yet challenges remain. Housing affordability for younger households has become a political concern as resale flat prices outpace wage growth. Healthcare and elderly care costs continue to rise. Some economists argue that headline GDP per capita, inflated by foreign professional incomes and corporate profits, overstates the typical citizen's standard of living.

Source: Adapted from The Straits Times, "Beyond the per-capita number", 4 February 2024.

Question 1

[4 marks]

With reference to Table 2.1, describe two changes in Singapore's living standards between 2010 and 2023.

Question 2

[4 marks]

Explain why a rise in nominal GDP per capita may overstate the improvement in the typical Singapore household's material standard of living.

Question 3

[4 marks]

Using evidence from Extract 2.1, explain how government redistributive policies have improved the standard of living in Singapore.

Question 4

[6 marks]

Explain why median household income is often a better indicator of the typical citizen's material standard of living than mean GDP per capita.

Question 5

[6 marks]

With reference to Table 2.1, explain how non-material factors have contributed to the improvement in Singapore's overall standard of living between 2010 and 2023.

Question 6

[6 marks]

Explain two reasons why, despite rising GDP per capita and falling income inequality, some Singapore households may still perceive their standard of living to have stagnated.

CASE STUDY 3. Beyond GDP: Alternative Measures of Living Standards

Table 3.1: Selected Country Comparison on Income and Alternative Living Standards Measures, 2022

Indicator	Norway	Singapore	Bhutan	Equatorial Guinea
GDP per capita, PPP US\$	114,900	127,600	11,800	18,200
HDI value (0–1)	0.966	0.949	0.681	0.654
HDI rank (out of 191)	2	12	125	145
Life Expectancy at Birth (years)	83.2	83.0	71.5	60.6
Mean Years of Schooling	13.0	11.9	5.2	6.2
Gini Coefficient (0–100)	27.7	37.1	37.4	—
World Happiness Index Rank	7	25	95	not ranked
GNH Index Score (Bhutan only)	—	—	0.781 (2022)	—

Source: Adapted from UNDP Human Development Report 2023, World Bank, World Happiness Report 2023, and the Centre for Bhutan Studies.

Extract 3.1: What do we really mean by standard of living?

Economists have long warned that GDP per capita captures only the monetary output produced within an economy. Equatorial Guinea, a small oil-exporting state in Central Africa, illustrates the extreme case. Its PPP GDP per capita of around 18,200 US dollars is higher than that of Bhutan, yet life expectancy is barely 60 years, mean years of schooling are only marginally higher, and the Human Development Index ranks Equatorial Guinea at 145th globally compared with Bhutan's 125th. Oil revenue has not translated into broad-based development.

The Human Development Index (HDI), introduced by the United Nations Development Programme in 1990, combines income, health, and education into a single composite score. Income is captured through the logarithm of GNI per capita, reflecting diminishing returns; health is captured through life expectancy at birth; education is captured through expected and mean years of schooling. By design, HDI rewards countries where growth has been broadly translated into longer, more educated lives.

Bhutan has taken the critique of GDP further. Since the 1970s, Bhutan has officially pursued Gross National Happiness (GNH) rather than GDP as its principal development objective. The GNH Index covers nine domains including psychological well-being, health, education, time use, cultural diversity, good governance, community vitality, ecological resilience, and living standards. The 2022 GNH Index recorded a score of 0.781, with 93.6 per cent of Bhutanese classified as “extensively happy” or “deeply happy”. Critics note that GNH is difficult to measure consistently and may not be transferable to larger or more heterogeneous societies. Supporters argue it captures dimensions of human flourishing that conventional GDP measures simply ignore.

Source: Adapted from The Economist, “Counting what counts”, 9 December 2023.

Question 1

[4 marks]

Using Table 3.1, explain why the case of Equatorial Guinea illustrates the limitations of GDP per capita as a measure of the standard of living.

Question 2

[4 marks]

Explain the three components of the Human Development Index and the rationale for including each.

Question 3

[4 marks]

Using Table 3.1, explain why Bhutan ranks higher on the Human Development Index than Equatorial Guinea, despite having a lower GDP per capita.

Question 4

[6 marks]

Explain one advantage and one limitation of using a composite index such as the Human Development Index instead of GDP per capita to compare living standards across countries.

Question 5

[6 marks]

Explain how Bhutan's Gross National Happiness framework attempts to address dimensions of the standard of living that conventional GDP measures fail to capture.

Question 6

[6 marks]

Explain two limitations of using the Gross National Happiness framework as a measure of the standard of living.

End of Booklet